Determinants of Health Icons

Biologic-Genetic

Behavioral

Healthcare

Social

Environmental

Clinical Presentation

The patient is John Swift. He is a 76-year-old man who presents to the ED with shortness of breath and palpitations.

Patient was mowing his lawn yesterday and started to experience chest pain, palpitations and SOB

Patient also said that they felt like their heart was racing and they were short of breath

He had to leave a party to rest because the pain he was experiencing was bad

Was brought to the ED by EMS because of his worsening symptoms

HPI

Patient is sweating as a result of the body activating the sympathetic nervous system activation to compensate for reduced cardiac output

Patient is also sweating and lightheaded

The patient is lightheaded because of reduced perfusion to the brain. When CO is reduced, the brain does not receive enough oxygen or blood

Patient is slightly tachycardic, but otherwise vitals were stable before I entered the room

Temperature: 98.9 F P: 115 RR: 20 BP: 110/70 O2: 98%

Patient may have a lack of knowledge regarding this which could potentially impact medication adherence

SDOH: Patient past medical history puts him at high risk for cardiovascular issues, especially Afib

PMH: Hypertension for the last 12 years, hyperlipidemia for the past 6 years and thyroid cancer for the past 4 years

Past Surgical History: Thyroidectomy

Allergies: NKA

Medications: Hydrochlorothiazide 25 mg daily, Aspirin 81 mg daily, Rosuvastatin 10 mg and levothyroxine 125 mg

Collect and understand patient information (History, Physical Exam, Testing)

Social history

Alcohol: 1-2 beers per day

SDOH: Alcohol use can also pose cardiovascular risks for patient

Recreational drugs: N/A

Tobacco: Smokes 0.5 packs a day and has a 20-year-pack history

SDOH: Smoking increases patients risk for cardiopulmonary disease and risk for further events of Afib

This provides a counseling opportunity to learn about patient and why they smoke

SDOH: Although I did not get the chance to speak with the patient, healthcare access and financial status can impact their health

If the patient does not follow-up regularly with a doctor or does not take their medications for financial reasons, these can all greatly impact his health

SDOH: Although I do not know the patient's living environment, his age can put him at risk for housing quality and adequate food access

This could be an opportunity to talk to patient about access and living condition concerns

SDOH: Lack of support or people to care for patient could also lead to problems in medication adherence or feelings of isolation or loneliness

SDOH: Based on living environment, pollution or other toxins could lead to issues for the patient on top of smoke inhalation from cigarettes which could exacerbate health issues

Physical exam

Patient had significant hypotension

Heart not pumping effectively because of loss of atrial kick which causes reduced ventricular filling which reduces CO and leads to hypotension

Patient was tachycardic

The sympathetic nervous system attempts to compensate for low perfusion, but the tachycardia can also contribute to reduced CO because the patient is in Afib

Crackles heard upon lung exam

The LV can not effectively pump which causes backflow to the lungs which causes increased hydrostatic pressure and leakage into the alveoli which leads to pulmonary edema and is heard as crackles

Peripheral edema was present

Elevated pressure can occur because of the backflow as well which causes increased hydrostatic pressure and leakage of the fluid into the interstitium

It stays in the interstitium because the lymphatic system cannot keep up with the amount of fluid

Diminished pulses

The reduced stroke volume of the heart leads to weak pulses

Patient was cool and clammy

The patient was cool and clammy because in cardiogenic shock, compensatory vasoconstriction occurs which leads to redirection of blood away from the skin to the vital organs

JVD was present

The backup from the failing heart can reach the jugular veins and pressure buildup causes that distention

Testing

EKG showed irregularly irregular heart rhythm consistent with atrial fibrillation

Chest X-ray showed pulmonary edema

Review of Systems

General: Anxious appearing, sweating, fatigue; no fever, chills, or weight loss

Respiratory: Shortness of breath; no cough

Cardiovascular: Palpitations and chest pain

Neurological: lightheadedness, no syncope or altered mental status

Endocrine: History of thyroidectomy; no current symptoms like weight changes mentioned

Pertinent positives: Irregularly irregular rhythm on EKG, tachycardia, palpitations, lightheadedness, SOB, HTN, age, pulmonary edema on chest X-ray ,JVD, and diminished pulses

Cardiogenic shock and Atrial fibrillation

Differential Diagnosis

Unstable angina/ NSTEMI

Exertional SOB, sweating, chest pain HTN, HLD, heavy smoking, aspirin use (indicating perceived CAD risk), and age are pertinent positives, but no elevated troponin and EKG was not indicative

Thyroxicosis

Post-thyroidectomy on levothyroxine, palpitations, tachycardia and sweating, but no recent medication change mentioned, no other symptoms like weight loss mentioned, EKG and pulmonary edema not consistent

Diagnosis Specific

Atrial fibrillation that led to cardiogenic shock

HSS Relation - Prevalence: 5.3-6.1 million Americans have AFib

HSS Relation - Incidence: There are about 1.2 million new cases of Afib per year

HSS: Atrial fibrillation occurs in about 28.2% of cardiogenic shock patients

Pathophysiology of Atrial Fibrillation and Cardiogenic shock

In atrial fibrillation, the atria contractions are disorganized which impacts atrial kick

As a result of the disorganized atria leading to tachycardia, preload decreases, stroke volume decreases, and as a result CO decreases

This decrease in cardiac output and the inability of the heart to compensate leads to very low CO which leads to hypotension and causes hypoperfusion of the organs

As a result of low CO, the sympathetic nervous system will kick in and cause vasoconstriction of blood vessels

The RAAS system also tries to compensation through fluid retention

However, this compensation actually worsens the condition

Vasoconstriction will increase afterload and the fluid will start to backup

The patient then experiences symptoms of fluid overload like peripheral edema, JVD, and pulmonary edema, and pulmonary edema can lead to the patient's SOB

Prevention, Diagnosis, Management

HSS Relation: Patient autonomy is very important. Ensure active listening with patient and understanding of patient point of view. Counsel patient with medical knowledge, but allow patient to have final say in their medical care.

Noted that patient was hypotensive, tachycardic, and tachypneic

I placed the patient on 100% non-rebreather which helped with the oxygen saturation

Also noted other signs of cardiogenic shock like JVD, peripheral edema, diminished pulses, pulmonary edema, crackles upon lung exam

Asked for EKG in which I saw that the patient was in atrial fibrillation

The patient was unstable so I decided to pursue synchronized cardioversion

After synchronized cardioversion, the patient was stable and feeling much better

After patient stabilized, I want to give furosemide to treat the pulmonary edema and fluid overload

I want to give heparin to prevent clotting because atrial fibrillation can cause clot formation

Monitor patient vitals

Counsel patient on smoking cessation and the risks it causes with his other comorbidities

Counsel patient on medication adherence for existing and new medications